Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ~~192.168.13.1~~ S0/0/0 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ~~110~~ 129

Does R2 show up as an OSPF neighbor on R1? **Yes** 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

R2's Serial0/0/1 interface is configured as a passive interface.
It does not send Hello packets, so it cannot form an OSPF neighbor relationship with R3.
However, R2 still advertises the192.168.2.0/24 network via ~~its other~~ interface. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

__ En config to router ospf 1 no passive-default s0/0/1 2 ________________________________

____

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

__~~110~~ 15 0 ____

Is R2 listed as an OSPF neighbor to R3? **Yes** 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To accurately calculate the cost of high-speed interfaces.The default reference bandwidth (100 Mbps) gives all interfaces faster than 100 Mbps a cost of 1, making them indistinguishable.Increasing the reference bandwidth allows OSPF to differentiate between Fast Ethernet, Gigabit Ethernet, and faster links, enabling more precise routing decisions.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24:

R1 S0/0/1 cost = 781 + R3 G0/0 cost = 1 → Total = 782

2

Cost to 192.168.23.0/30:

R1 S0/0/1 cost = 781 + R3 S0/0/1 cost = 64 → Total = 845

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562 Because the route now goes through two serial links:

R1 S0/0/0 cost = 781, R2 S0/0/1 cost = 781, Total = 781 × 2 = 1562 2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF always selects the path with the lowest cumulative cost.

4

Via R1 S0/0/1 (cost 1565) + R3 G0/0 (cost 1) = 1566, Via R1 S0/0/0 (cost 781) + R2 S0/0/1 (cost 781) + R3 G0/0 (cost 1) = 1563,Since 1563 < 1566, OSPF chooses the path through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   To ensure stability and predictability. 4

   If the router ID is dynamically chosen from an active interface, it may change if that interface goes down, disrupting OSPF operations.

   Manually setting the router ID prevents this and ensures consistent neighbor relationships.

2. Why is the DR/BDR election process not a concern in this lab? 4

   DR/BDR election only occurs on multi-access networks (e.g., Ethernet).
   All serial links in this lab are point-to-point, so no DR/BDR election is needed.

3. Why would you want to set an OSPF interface to passive?

   To reduce unnecessary routing traffic, improve security, and save bandwidth. 4
   On LAN interfaces where no OSPF neighbors exist, setting them as passive prevents Hello packets from being sent while still advertising the connected network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of **areas**.
   OSPF divides the network into a backbone area (Area 0) and other areas, reducing the size of link-state databases and limiting the impact of topology changes. 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   Router(config-if)#ip address 10.0.0.0 255.255.255.0 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   RIP, OSPF and EIGRP. EIGRP 0